Types of APIs

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What Are APIs ?

APIs (Application Programming Interfaces) are tools that allow different software systems to communicate with each other, enabling them to share data and functionality.

Why are APIs important?

- They help apps and websites share data and features.
- APIs make development faster by connecting services.

We'll look at 4 common types of APIs:

- REST API
- SOAP API
- GraphQL API
- WebSocket API

1) REST API

(Representational State Transfer)

What is it?

A widely used web architecture for communication over HTTP.

When to use it:

- When you need a simple, stateless API that works well with HTTP methods.
- Best for CRUD (Create, Read, Update, Delete) operations.

2) SOAP API

(Simple Object Access Protocol)

What is it?

A protocol for exchanging structured information, often used in enterprise apps.

When to use it:

- When strict standards and high security are required.
- Ideal for financial services, payment gateways, or any enterprise-level applications.

script.js

```
const soap = require('soap');
const url = 'https://example.com/service?wsdl';
soap.createClient(url, (err, client) => {
  client.MyFunction(args, (err, result) => {
    console.log(result);
  });
});
```

3) GraphQL API

What is it?

A flexible query language for APIs that allows clients to request only the data they need.

When to use it:

- When you need more control over the data you fetch (e.g., avoiding over-fetching).
- Great for applications with complex data relationships.

```
script.js

const query = `
  query {
    posts {
      id
      title
      author {
        name
      }
    }
  }
`;

fetch('https://example.com/graphql', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ query })
})
  .then(response => response.json())
  .then(data => console.log(data));
```

4) WebSocket API

What is it?

Enables real-time, two-way communication between client and server over a persistent connection.

When to use it:

- When you need real-time communication, like in chat applications or live updates.
- Ideal for applications that require low latency and continuous data flow.

script.js

```
const socket = new WebSocket('wss://example.com/socket');
socket.onopen = () => socket.send('Hello Server');
socket.onmessage = (event) => console.log('Message from server:', event.data);
```

RESTful vs. SOAP vs. GraphQL -Which One to Choose?

REST API

- Best for most web applications and services.
- Simple, flexible, and widely supported.

SOAP API

- Ideal for legacy enterprise systems with strict standards.
- Requires more setup but offers more security and reliability.

GraphQL API

- Best for complex data fetching needs, allowing clients to specify exactly what they need.
- Perfect for applications with evolving data structures or dynamic data sources.

Key Factors in Choosing the Right API

1. Data Complexity:

- REST is simple; GraphQL is more flexible for complex data.
- SOAP is best for strict, predefined structures.

2. Real-Time Needs:

- WebSocket is ideal for real-time, two-way communication.

3. Security:

- SOAP offers higher security, especially in financial or enterprise applications.

4. Scalability:

- REST APIs are highly scalable, making them a go-to choice for most web apps.

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